Green Tea Polyphenol (-)-Epicatechin Pretreatment Mitigates Hepatic Steatosis in an In Vitro MASLD Model.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously known as non-alcoholic fatty liver disease (NAFLD), is becoming more prominent globally due to an increase in the prevalence of obesity, dyslipidemia, and type 2 diabetes. A great deal of studies have proposed potential treatments for MASLD, with few of them demonstrating promising results. The aim of this study was to investigate the potential effects of (-)-epicatechin (EPI) on the development of MASLD in an in vitro model using the HepG2 cell line by determining the metabolic viability of the cells and the levels of PPARα, PPARγ, and GSH. HepG2 cells were pretreated with 10, 30, 50, and 100 μM EPI for 4 h to assess the potential effects of EPI on lipid metabolism. A MASLD cell culture model was established using HepG2 hepatocytes which were exposed to 1.5 mM oleic acid (OA) for 24 h. Moreover, colorimetric MTS assay was used in order to determine the metabolic viability of the cells, PPARα and PPARγ protein levels were determined using enzyme-linked immunosorbent assay (ELISA), and lipid accumulation was visualized using the Oil Red O Staining method. Also, the levels of intracellular glutathione (GSH) were measured to determine the level of oxidative stress. EPI was shown to increase the metabolic viability of the cells treated with OA. The metabolic viability of HepG2 cells, after 24 h incubation with OA, was significantly decreased, with a metabolic viability of 71%, compared to the cells pretreated with EPI, where the metabolic viability was 74-86% with respect to the concentration of EPI used in the experiment. Furthermore, the levels of PPARα, PPARγ, and GSH exhibited a decrease in response to increasing EPI concentrations. Pretreatment with EPI has demonstrated a great effect on the levels of PPARα, PPARγ, and GSH in vitro. Therefore, considering that EPI mediates lipid metabolism in MASLD, it should be considered a promising hepatoprotective agent in future research.

Diagnosis and management of metabolic dysfunction- associated steatotic liver disease in South Asians- A clinical review.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously termed as nonalcoholic fatty liver disease (NAFLD) is a hepatic manifestation of obesity and metabolic syndrome. It is mainly caused by insulin resistance. With the increased risk of visceral obesity in South Asians, the prevalence of MASLD is on the rise. The morbidity associated with MASLD and its complications, including hepatocellular carcinoma is projected to increase in this South Asian population. In this narrative review we explore the diagnosis and management of MASLD in the South Asian population. We summarize the findings from the recent literature on the diagnostic methods and management options for MASLD in this population. Through our search we found no specific guidelines for the diagnosis and management of MASLD in the South Asian population. The existing general guidelines may not be applied to South Asian populations due to the differences in phenotype, genotype, social and cultural aspects. South Asian countries also have limited resources with the non-availability of newer pharmacotherapeutic agents. The goal of this review is to guide obesity physicians and primary care providers to have a stepwise approach to treat patients at risk for MASLD with a main focus on interdisciplinary management most applicable to South Asian patients. More research is needed to formulate guidelines and algorithm that are specific for the South Asian population.

Metabolic dysfunction-associated steatotic liver disease and cardiovascular risk: a comprehensive review.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously termed nonalcoholic fatty liver disease (NAFLD), poses a significant global health challenge due to its increasing prevalence and strong association with cardiovascular disease (CVD). This comprehensive review summarizes the current knowledge on the MASLD-CVD relationship, compares analysis of how different terminologies for fatty liver disease affect cardiovascular (CV) risk assessment using different diagnostic criteria, explores the pathophysiological mechanisms connecting MASLD to CVD, the influence of MASLD on traditional CV risk factors, the role of noninvasive imaging techniques and biomarkers in the assessment of CV risk in patients with MASLD, and the implications for clinical management and prevention strategies. By incorporating current research and clinical guidelines, this review provides a comprehensive overview of the complex interplay between MASLD and cardiovascular health.

Pathology and Pathogenesis of Metabolic Dysfunction-Associated Steatotic Liver Disease-Associated Hepatic Tumors.

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Nonalcoholic fatty liver disease (NAFLD) is characterized by excessive fat accumulation in the livers of patients without a history of alcohol abuse. It is classified as either simple steatosis (nonalcoholic fatty liver) or nonalcoholic steatohepatitis (NASH), which can progress to liver cirrhosis and hepatocellular carcinoma (HCC). Recently, it was suggested that the terms "metabolic dysfunction-associated steatotic liver disease (MASLD)" and "metabolic dysfunction-associated steatohepatitis (MASH)" should replace the terms "nonalcoholic fatty liver disease (NAFLD)" and "nonalcoholic steatohepatitis (NASH)", respectively, with small changes in the definitions. MASLD, a hepatic manifestation of metabolic syndrome, is rapidly increasing in incidence globally, and is becoming an increasingly important cause of HCC. Steatohepatitic HCC, a histological variant of HCC, is characterized by its morphological features resembling non-neoplastic steatohepatitis and is closely associated with underlying steatohepatitis and metabolic syndrome. Variations in genes including patatin-like phospholipase domain-containing protein 3 (PNPLA3), transmembrane 6 superfamily 2 (TM6SF2), and membrane-bound O-acyltransferase domain-containing protein 7 (MBOAT7) are associated with the natural history of MASLD, including HCC development. The mechanisms of HCC development in MASLD have not been fully elucidated; however, various factors, including lipotoxicity, inflammation, reactive oxygen species, insulin resistance, and alterations in the gut bacterial flora, are important in the pathogenesis of MASLD-associated HCC. Obesity and MASLD are also recognized as risk factors for hepatocellular adenomas, and recent meta-analyses have shown an association between MASLD and intrahepatic cholangiocarcinoma. In this review, we outline the pathology and pathogenesis of MASLD-associated liver tumors.

NAD metabolic therapy in metabolic dysfunction-associated steatotic liver disease: Possible roles of gut microbiota.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly named non-alcoholic fatty liver disease (NAFLD), is induced by alterations of hepatic metabolism. As a critical metabolites function regulator, nicotinamide adenine dinucleotide (NAD) nowadays has been validated to be effective in the treatment of diet-induced murine model of MASLD. Additionally, gut microbiota has been reported to have the potential to prevent MASLD by dietary NAD precursors metabolizing together with mammals. However, the underlying mechanism remains unclear. In this review, we hypothesized that NAD enhancing mitochondrial activity might reshape a specific microbiota signature, and improve MASLD progression demonstrated by fecal microbiota transplantation. Here, this review especially focused on the mechanism of Microbiota-Gut-Liver Axis together with NAD metabolism for the MASLD progress. Notably, we found significant changes in Prevotella associated with NAD in a gut microbiome signature of certain MASLD patients. With the recent researches, we also inferred that Prevotella can not only regulate the level of NAD pool by boosting the carbon metabolism, but also play a vital part in regulating the branched-chain amino acid (BCAA)-related fatty acid metabolism pathway. Altogether, our results support the notion that the gut microbiota contribute to the dietary NAD precursors metabolism in MASLD development and the dietary NAD precursors together with certain gut microbiota may be a preventive or therapeutic strategy in MASLD management.

The impact of published guidance on trends in the pharmacological management of depression in children and adolescents- a whole population e-cohort data linkage study in Wales, UK.

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This study evaluated the impact of 2015/2016 prescribing guidance on antidepressant prescribing choices in children. A retrospective e-cohort study of whole population routine electronic healthcare records was conducted. Poisson regression was undertaken to explore trends over time for depression, antidepressant prescribing, indications and secondary care contacts. Time trend analysis was conducted to assess the impact of guidance. A total of 643 322 primary care patients in Wales UK, aged 6-17 years from 2010-2019 contributed 3 215 584 person-years of follow-up. Adjusted incidence of depression more than doubled (IRR for 2019 = 2.8 [2.5-3.2]) with similar trends seen for antidepressants. Fluoxetine was the most frequently prescribed first-line antidepressant. Citalopram comprised less than 5% of first prescriptions in younger children but 22.9% (95% CI 22.0-23.8; 95% CI 2533) in 16-17-year-olds. Approximately half of new antidepressant prescribing was associated with depression. Segmented regression analysis showed that prescriptions of 'all' antidepressants, Fluoxetine and Sertraline were increasing before the guidance. This upward trend flattened for both 'all' antidepressants and Fluoxetine and steepened for Sertraline. Citalopram prescribing was decreasing significantly pre guidance being issued with no significant change afterward. Targeted intervention is needed to address rising rates of depression in children. Practitioners are partially adhering to local and national guidance. The decision-making process behind prescribing choices is likely to be multi-factorial. Activities to support implementation of guidance should be adopted in relation to safety in prescribing of antidepressants in children including timely availability of talking therapies and specialist mental health services.

Temporal trends of antidepressant utilization patterns in children and adolescents in Hong Kong: A 14-year population-based study with joinpoint regression analysis.

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There is limited research on real-world antidepressant utilization patterns in children and adolescents, particularly in non-western countries. We aimed to examine temporal trends of antidepressant prescribing practice among Chinese children and adolescents in Hong Kong over 14-year period. This population-based study identified 9566 patients aged 5-17 years who had redeemed at least one antidepressant prescription within 2005-2018, using data from health-record database of Hong Kong public healthcare services. We calculated annual prescription rates (per 1000 persons) for any antidepressant, antidepressant drug classes, and individual antidepressants. Joinpoint-regression analyses were performed to assess temporal antidepressant prescription trends, quantified by average annual-percent-change (AAPC), with 95 % confidence-intervals (CIs). Overall antidepressant prescription rate significantly increased over time (AAPC: 7.30 [95 % CI: 6.70-7.90]), from 3.883 in 2005 to 9.916 in 2018. The use of selective-serotonin-reuptake-inhibitors (SSRIs), serotonin-norepinephrine-reuptake-inhibitors (SNRIs), and other antidepressants significantly increased over 14 years, while tricyclic-antidepressants remained stable. SSRI represented the most commonly-prescribed drug class. Fluoxetine and sertraline constituted the two most frequently-prescribed individual antidepressants, while desvenlafaxine (AAPC: 55.68 [30.74-85.39]) and bupropion (AAPC: 35.28 [23.68-47.98]) exhibited the sharpest increase in prescription rates over the study period. Medication adherence could not be assessed and actual drug use may be overestimated. Our results affirm a significant rising trend of antidepressant prescriptions among Chinese children and adolescents over time. All antidepressant drug-classes, except TCA, demonstrated significantly increased use, with SSRI being the most frequently-prescribed drug class. Future investigation should clarify indications, hence off-label use, of antidepressant initiation in this vulnerable population.

Adverse Effects of Antidepressant Medications and their Management in Children and Adolescents.

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Selective serotonin reuptake inhibitors (SSRIs) and, to a lesser extent, serotonin-norepinephrine reuptake inhibitors (SNRIs) are the cornerstone of pharmacotherapy for children and adolescents with anxiety and depressive disorders. These medications alleviate symptoms and restore function for many youths; however, they are associated with a distinct adverse effect profile, and their tolerability may complicate treatment or lead to discontinuation. Yet, SSRI/SNRI tolerability has received limited attention in the pediatric literature. This review examines the early- (e.g., activation, gastrointestinal symptoms, sedation) and late-emerging (e.g., weight gain) adverse effects of SSRIs and some SNRIs in pediatric patients. We provide a framework for discussing SSRI/SNRI tolerability with patients and their families and describe the pharmacologic basis, course, and predictors of adverse events in youth. Strategies to address specific tolerability concerns are presented. For selected adverse events, using posterior simulation of mean differences over time, we describe their course based on Physical Symptom Checklist measures in a prospective, randomized trial of anxious youth aged 7-17 years who were treated with sertraline ($n = 139$) or placebo ($n = 76$) for 12 weeks in the Child/Adolescent Anxiety Multimodal Study (CAMS). In CAMS, the relative severity/burden of total physical symptoms ($p < 0.001$), insomnia ($p = 0.001$), restlessness ($p < 0.001$), nausea ($p = 0.002$), abdominal pain ($p < 0.001$), and dry mouth ($p = 0.024$) decreased from baseline over 12 weeks of sertraline treatment, raising the possibility that these symptoms are transient. No significant changes were observed for sweating ($p = 0.103$), constipation ($p = 0.241$), or diarrhea ($p = 0.489$). Finally, we review the antidepressant withdrawal syndrome in children and adolescents and provide guidance for SSRI discontinuation, using pediatric pharmacokinetic models of escitalopram and sertraline-two of the most used SSRIs in youth. SSRI/SNRIs are associated with both early-emerging (often transient) and late-emerging adverse effects in youth. Pharmacokinetically-informed appraoches may address some adverse effects and inform SSRI/SNRI discontinuation strategies.

Long-term sertraline treatment of children and adolescents with major depressive disorder.

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The aim of this study was to assess the long-term safety, tolerability, and efficacy of sertraline 50-200 mg once-daily in children (6-11 year olds) and adolescents (12-18 year olds) with a Diagnostic and Statistical Manual of Mental Disorders, 4th edition (DSM-IV) diagnosis of major depressive disorder (MDD). This study consisted of a 24-week open-label observational study of children and adolescents who had completed either of two 10-week double-blind, placebo-controlled trials. The Children's Depression Rating Scale-Revised (CDRS-R) was the primary measure of efficacy. Two hundred ninety nine (299) patients completed the acute studies and were eligible for the extension study. Of these, 226 enrolled, but 5 did not receive treatment. Of 221 patients (107 children and 114 adolescents), 62.4% completed the study. The endpoint mean daily dose was 109.9 mg/day. The mean decrease in CDRS-R score from double-blind baseline was 34.8 points ($p < 0.001$), with patients showing continued improvement in CDRS-R scores regardless of which treatment they received in the double-blind studies. At endpoint, 86% of patients met CDRS-R responder and 58% CDRS-R remitter criteria. Sertraline appears to be well tolerated and safe over 24 weeks of treatment in children and adolescents with MDD. Children and adolescents treated with sertraline appear to have increased improvement over that seen in the first 10 weeks of treatment. These findings need confirmation in placebo-controlled studies.

Sertraline: new indication. May help children with obsessive-compulsive disorder.

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(1) The choice of treatment for children with obsessive-compulsive disorder is difficult. Behaviour therapy and antidepressants have not been assessed adequately in this setting, and their efficacy seems limited. Clomipramine was the first antidepressant to show a degree of efficacy. (2) Sertraline is the first drug to be licensed in France for children aged from 6 to 17 years with obsessive-compulsive disorder. (3) According to our literature search, the evaluation file on sertraline in this indication mainly contains data from a double-blind placebo-controlled trial involving 187 children. After 3 months of treatment, sertraline was significantly more effective than placebo, although most children remained symptomatic. Direct comparison is lacking, but sertraline seems as effective as clomipramine. (4) However, 13% of children receiving sertraline left this trial because of adverse events (3% on placebo; $p = 0.02$). The short-term safety profile of sertraline in children is the same as in adults, i.e. mainly nausea, agitation, headache, insomnia and tremor. (5) We have no data on the effects of prolonged sertraline therapy in children, particularly on neuropsychological development. (6) The first-line treatment of obsessive-compulsive disorder is behaviour therapy. Sertraline, like clomipramine, is an option when behaviour therapy fails or is unfeasible. The choice between sertraline and clomipramine should be

discussed case by case, according to their safety profiles; however, we have more experience with clomipramine, which should therefore be preferred over sertraline.